

10.2 | Kirchhoff's Laws

- ① Sum of current into junction
= Sum of current out of junction

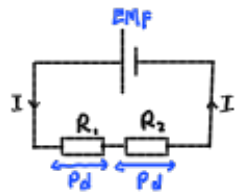
(OR)

Algebraic sum of current
at junction = 0

$$\sum I_{in} = \sum I_{out}$$

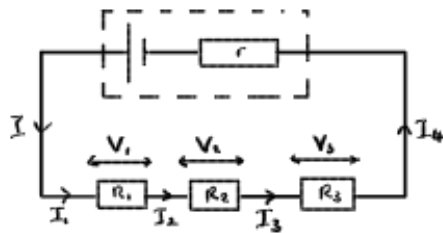
- ② Sum of EMF = Sum of Pd

$$\sum EMF = \sum Pd$$



$$\mathcal{E} = IR_1 + IR_2$$

Eg 1: Resistors in series.



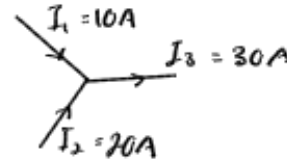
$$\textcircled{1} I = I_1 = I_2 = I_3 = I_4$$

$$\textcircled{2} \mathcal{E} = V_1 + V_2 + V_3 + V_r$$

$$IR_{Total} = I_1 R_1 + I_2 R_2 + I_3 R_3 + I_4 r$$

$$\textcircled{3} R_T = R_1 + R_2 + R_3 + r$$

10.3 | Potential dividers



$$\uparrow V_1 = I_1 R_1 \text{ same}$$

$$\uparrow \downarrow V_2 = I_2 R_2$$

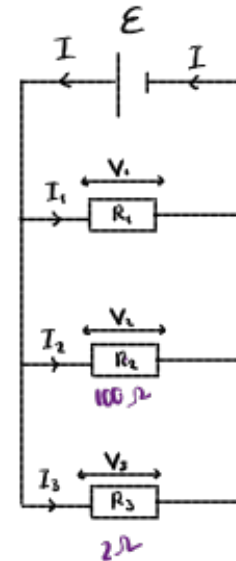
$$\uparrow \downarrow V_T = I_T R_T$$

$$V_b = IR_b$$

$$4.3 = IR_b$$

$$5 = I[2 + R_b]$$

Eg 2: Resistors in parallel



$$\textcircled{1} I = I_1 + I_2 + I_3$$

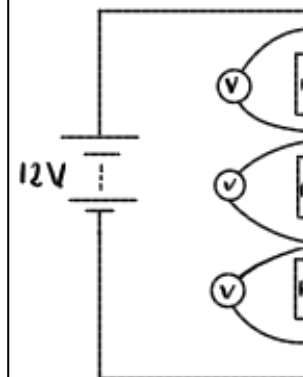
$$\textcircled{2} \mathcal{E} = V_1 = V_2 = V_3$$

$$\textcircled{3} \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$V_1 = IR_1$$

$$\text{constant } V = \frac{\downarrow \uparrow}{I} R$$

Eg 3: Potential Divider



$$\textcircled{1} \sum I_{in} = \sum I_{out}$$

$$\textcircled{2} \sum \mathcal{E} = \sum pd$$

$$\textcircled{1} I = I_1 = I_2 = I_3$$

$$\frac{\mathcal{E}}{R_T} = \frac{V_1}{R_1} = \frac{V_2}{R_2} = \frac{V_3}{R_3}$$

$$V_x = R_x(I) = R_x\left(\frac{\mathcal{E}}{R_T}\right)$$

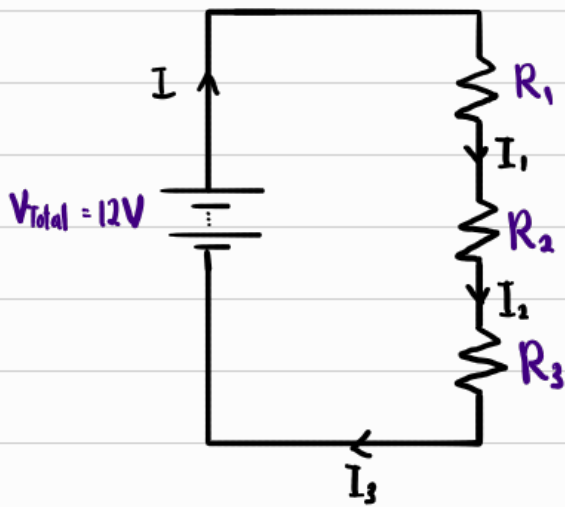
$$\textcircled{2} V_x = \left(\frac{R_x}{R_T}\right) \mathcal{E}$$

$$I_{in} = I_{out} \text{ [in series]}$$

$$\frac{V_{in}}{R_T} = \frac{V_{out}}{R_{T_{out}}}$$

$$V_{out} = V_{in} \times \left(\frac{R_{T_{out}}}{R_T}\right)$$

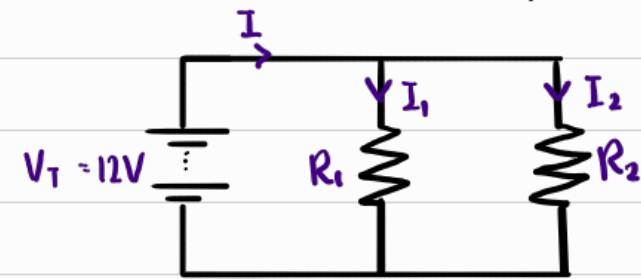
Potential Divider Concept



① $I = I_1 = I_2 = I_3$ [In series]
 $\Rightarrow \frac{V_T}{R_T} = \frac{V_{R_1}}{R_1} = \frac{V_{R_2}}{R_2} = \frac{V_{R_3}}{R_3}$

$\therefore \frac{V_{in}}{R_T} = \frac{V_{out}}{R_{Tout}}$

Current Divider Concept



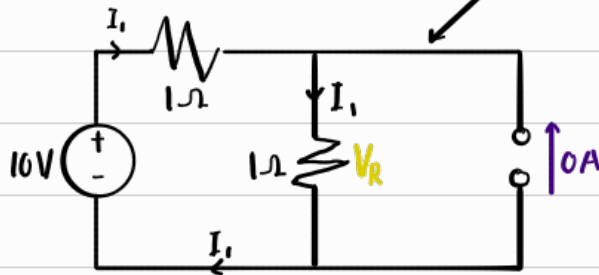
① $V_T = V_1 = V_2$ [In parallel]
 $IR_T = I_1 R_1 = I_2 R_2$

$\therefore I_1 = \frac{R_T}{R_1} I$

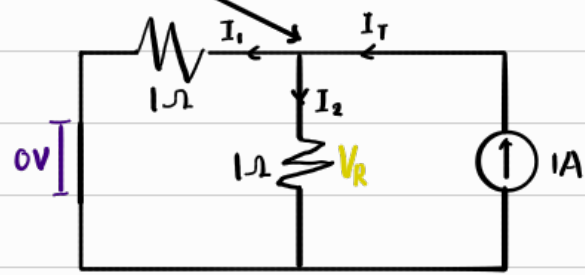
Super position principle



Set 1A source
to 0A
(Open circuit)



Set 10V source
to 0V
(Short circuit)



Since $I_1 = I_2$

$$\frac{V_R^{(10V)}}{1} = \frac{V_{10V}}{R_T}$$

$$V_R^{(10V)} = \frac{10}{1+1} (1)$$

Pd at R caused
by 10V source = 5V

Resistor are in parallel,

$$\Rightarrow \frac{1}{R_T} = \frac{1}{1} + \frac{1}{1}$$

$$R_T = 0.5\Omega$$

$$V_{1A} = I_T R_T$$

$$= 1 (0.5)$$

$$= 0.5V$$

Since 1A source is parallel w V_R ,

$$V_R^{(1A)} = V_{1A}$$

$$= 0.5V$$

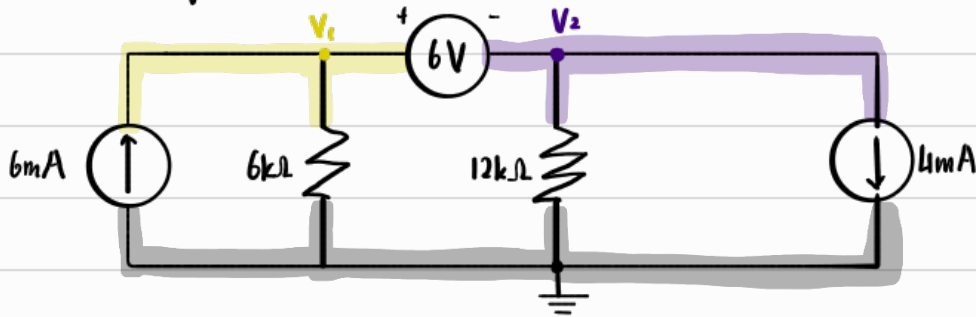
$$\therefore V_R = V_R^{(10\Omega)} + V_R^{(1A)}$$

$$= 5 + 0.5$$

$$V_R = 5.5V$$

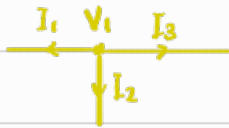
* Only Independent sources can be set to zero

Nodal Analysis



Looking at node V_1

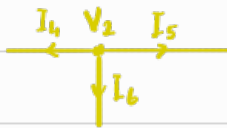
* Assume +ve is away from node



From KCL, we know that $I_1 + I_2 + I_3 = 0$

$$\Rightarrow -6\text{mA} + \frac{V_1 - 0}{6\text{k}\Omega} + I_3 = 0 \quad \text{--- (1)}$$

Looking at node V_2



From KCL, we know that $I_4 + I_5 + I_6 = 0$

$$\Rightarrow -I_3 + \frac{V_2 - 0}{12\text{k}\Omega} + 4\text{mA} = 0, \quad [I_4 = -I_3] \quad \text{--- (2)}$$

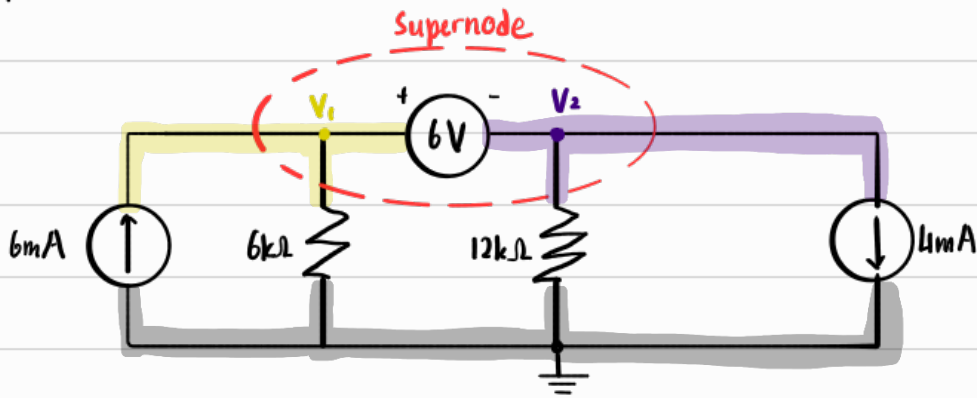
From voltage source definition

$$V_1 - V_2 = 6\text{V} = 0, \quad [\text{Beuz from } V_2 \text{ to } V_1, \text{ there is a gain of } 6\text{V}] \quad \text{--- (3)}$$

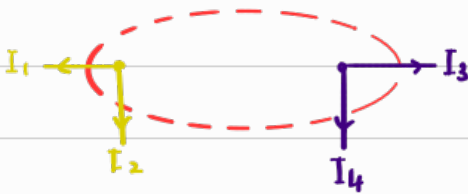
Now solve simul. to get V_1 , V_2 and I_3 .

[3 eqn, 3 unknown]

Supernode [Shortcut]



Looking at supernode



From KCL, we know that $I_1 + I_2 + I_3 + I_4 = 0$

$$\Rightarrow -6\text{mA} + \frac{V_1 - 0}{6\text{k}\Omega} + \frac{V_2 - 0}{12\text{k}\Omega} + 4\text{mA} = 0 \quad \text{--- (1)}$$

From voltage source definition

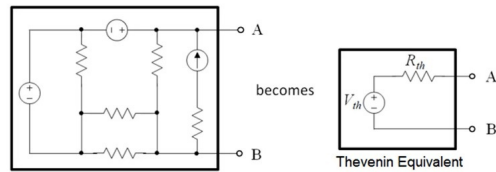
$$V_1 - V_2 = 6\text{V} = 0, \text{ [Beuz from } V_2 \text{ to } V_1, \text{ there is a gain of } 6\text{V}] \quad \text{--- (2)}$$

Now solve simul. to get V_1 and V_2

[2 eqn, 2 unknowns]

Thevenin Theorem and Norton Theorem

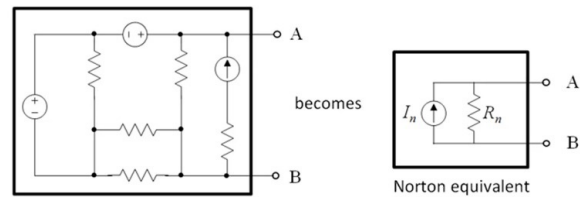
- Thevenin's Theorem states that it is possible to simplify any linear circuit relative to two terminals, no matter how complex, to an equivalent circuit with just a single voltage source and series resistance.



- V_{th} is called the Thevenin voltage
- R_{th} is called the Thevenin resistance



- Norton's Theorem states that it is possible to simplify any linear circuit relative to two terminals, no matter how complex, to an equivalent circuit with just a single current source and parallel resistance.



- I_n is called the Norton current
- R_n is called the Norton resistance



Essentially, we're just simplifying the circuit in the perspective of the load. So that when we change the load, we can easily calculate its V, I, R without needing to recalculate the entire circuit again.

In the POV of the load, there is only 1 voltage drop across it, and one current through it. The magnitude of the current is determined by the equivalence Resistance

$$V_{oc} = I_{sc} R_{eq}$$

Open circuit voltage = Short circuit voltage $\times$ Equivalence Resistance

When Calculating Thevenin Resistance (R_{th}):

- Turn off all independent sources:
 - Voltage sources $\rightarrow$ Replace with a short circuit.
 - Current sources $\rightarrow$ Replace with an open circuit.
- Remove the load resistor.
- Calculate the equivalent resistance seen from the load terminals. This is the Thevenin resistance.

When Calculating Thevenin Voltage (V_{th}):

- Leave all sources active (don't turn off anything).
- Remove the load resistor.
- Find the open-circuit voltage across the load terminals. This is the Thevenin voltage.

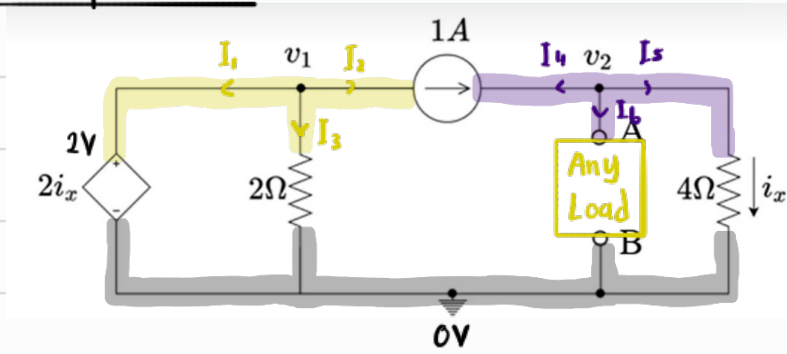
Norton Resistance (R_N):

- Turn off all independent sources (just like with Thevenin):
 - Voltage sources $\rightarrow$ Replace with a short circuit.
 - Current sources $\rightarrow$ Replace with an open circuit.
- Remove the load resistor.
- Calculate the equivalent resistance seen from the load terminals. This is the Norton resistance, which is the same as the Thevenin resistance ($R_{th} = R_N$).

Norton Current (I_N):

- Leave all sources active (don't turn off anything).
- Short the load terminals (just like for Thevenin current).
- Calculate the current that flows through the shorted load terminals. This is the Norton current, which is the same as the Thevenin current in magnitude, but it's a current source instead of a voltage source.

Example 1:



① Find V_{oc} [Assume AB is a open circuit, 0A]

$$V_1 = 2i_x \quad \text{--- ①}$$

Node V_2

$$\text{Using KCL, } I_4 + I_5 + I_6 = 0$$

$$\Rightarrow -1A + i_x + 0 = 0$$

$$i_x = 1A \quad \Rightarrow V_1 = 2(1) = 2V \quad \#$$

$$\frac{V_2 - 0}{4} = 1$$

$$V_2 = 4V \quad \#$$

$$\therefore V_{oc} = V_2 = 4V \quad \#$$

② Find I_{sc} [Assume AB is a short circuit, 0V]

$$i_x = 0 \quad [\text{Beuz AB is a short circuit, ALL current flow thru AB}]$$

$$V_1 = 2i_x = 0 \quad \#$$

$$V_2 - 0 = 0$$

$$V_2 = 0 \quad \#$$

Node V_2

$$\text{Using KCL, } I_4 + I_5 + I_6 = 0$$

$$\Rightarrow -1A + I_{sc} + i_x = 0$$

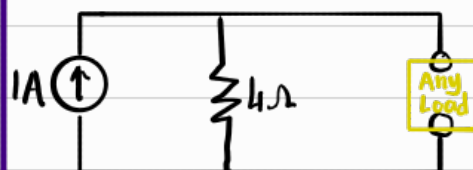
$$I_{sc} = 1A \quad \#$$

$$\therefore R_{eq} = \frac{V_{oc}}{I_{sc}} = \frac{4}{1} = 4\Omega$$

Thevenin



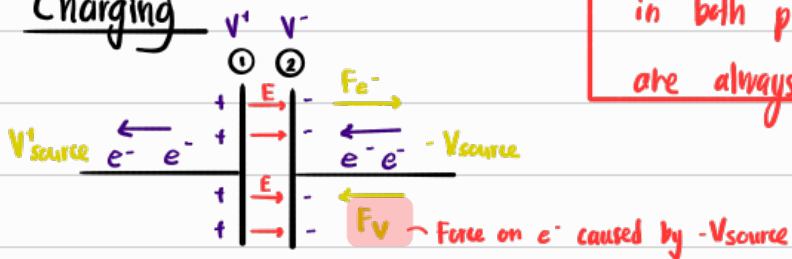
Norton



Capacitors

Working Principle

Charging



* Since when $1 e^-$ in plate ② repels $1 e^-$ from plate ①, the magnitude of charges is the same in both plates but diff polarity. Thus capacitors are always neutral as a whole

- ① As the $-V$ pushes e^- onto the originally neutral plate ②, plate ② starts getting more and more $-ve$

[This causes e^- to repel against each other, producing a force on the e^- to move back]

- ② However, the net $-ve$ charge of plate ② repels e^- from the originally neutral plate ①

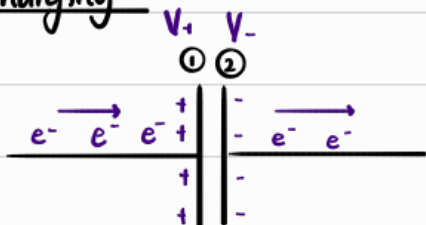
- ③ This causes an E -field that attracts e^- on plate ② to avoid e^- on plate ② repelling each other and moving backwards

- ④ Process ① happens until " $E + F_v = F_{e^-}$ " OR $V_{capacitor} = V_{source}$

Since $pd = 0$, $I = 0$, e^- won't move anymore

[E can't keep increasing becu the no. of $+ve$ charges on plate ① is limited]

Discharging



- ① When $V_{capacitor} > V_{source}$, e^- will start going into $+ve$ plate ①

- ② This reduces E -field, thus causing " $E + F_v < F_{e^-}$ "

- ③ Due to ②, e^- on plate ② repels each other to the right

- ④ Process ① & ③ causes $V_{capacitor}$ and $V_{capacitor}$ to reduce

- ⑤ Process ① & ③ stops when both plates become neutral again, causing

$V_{capacitor}$ & $V_{capacitor}$ to be zero.

$pd = 0$, $I = 0$, no movement of charges.

Capacitance, C

$$C = Q/V$$

Capacitance is the no. of charges the capacitor can hold per volt

Factors affecting capacitance

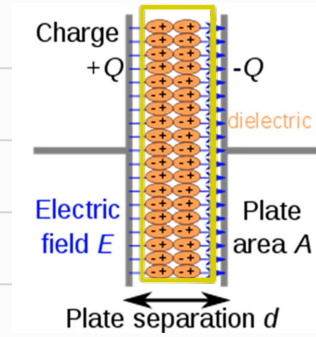
① Type of insulator

for example, dielectric insulator.

- The charges in the dielectric insulator arranges itself to attract the e^- on the -ve plate

↳ More e^- can be accumulated on the -ve plate without being repelled backwards, $Q \uparrow$

↳ $C \uparrow$



② Plate separation, d

- As $d \downarrow$, $E \uparrow$

↳ More e^- can be accumulated on the -ve plate without being repelled backwards, $Q \uparrow$

↳ $C \uparrow$

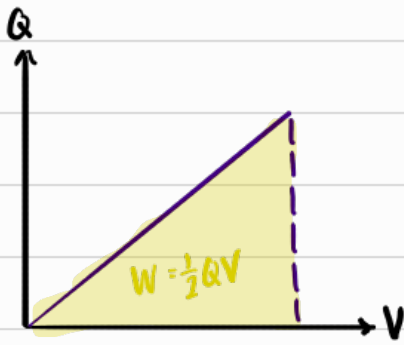
③ Area of parallel plates, A

- As $A \uparrow$, the plate have more space to thus same e^- repulsion accomodate more e^- at the same charge density becu more e^- can fit while keeping the distance btwn e^- the same

↳ $Q \uparrow$

↳ $C \uparrow$

Characteristics



$$\therefore \text{Energy stored in capacitor, } E = \frac{1}{2} QV \text{ --- (1)}$$

$$V = Q/c \text{ --- (2)}$$

$$C = \frac{Q}{V} \quad \left\{ \begin{array}{l} V = Q/c \\ Q = CV \end{array} \right. \text{ --- (3)}$$

$$\text{(2)} \rightarrow \text{(1)}$$

$$E = \frac{1}{2} \left(Q^2 / c \right)$$

$$\text{(2)} \rightarrow \text{(1)}$$

$$E = \frac{1}{2} CV^2$$

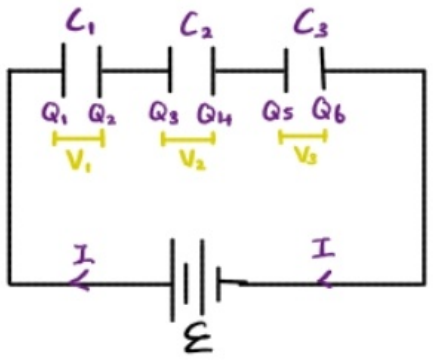
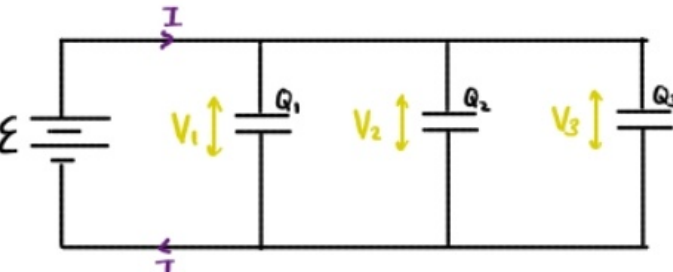
$$I_c(t) = \frac{dQ}{dt} \\ = \frac{d(CV(t))}{dt}$$

$$I_c(t) = C \left[\frac{dV(t)}{dt} \right]$$

inversely $\rightarrow$

$$V_c(t) = \left[\frac{1}{C} \int_{t_0}^t I_c(\tau) d\tau \right] + V_c(t_0)$$

4. Combined Capacitance: Capacitors in Series & Parellel

Series Arrangement	Parallel Arrangement
	
<u>Charge</u> $Q_1 = Q_3 = Q_5 \quad \left. \begin{array}{l} \text{Same magnitude} \\ \text{Opposite Dirrec.} \end{array} \right\}$ $Q_2 = Q_4 = Q_6$ * Charge stored by each capacitor is the same	$Q_{\text{supply}} = Q_1 + Q_2 + Q_3$
<u>Voltage</u> $\mathcal{E} = V_1 + V_2 + V_3$	$\mathcal{E} = V_1 = V_2 = V_3$
<u>Combined capacitance</u> $\mathcal{E} = V_1 + V_2 + V_3 \quad \boxed{V = \frac{Q}{C}}$ $\frac{Q_{\text{supply}}}{C_{\text{Total}}} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$ $\frac{Q_{\text{supply}}}{C_{\text{Total}}} = Q_{\text{supply}} \left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right]$ $\boxed{\frac{1}{C_{\text{Total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$	$Q_{\text{supply}} = Q_1 + Q_2 + Q_3$ $(\mathcal{E})(C_T) = V_1 C_1 + V_2 C_2 + V_3 C_3$ $(\mathcal{E})(C_T) = \mathcal{E} [C_1 + C_2 + C_3]$ $\boxed{C_T = C_1 + C_2 + C_3}$

Resistance & Capacitor circuits [RC circuits]

General solution : $V_c(t) = K_1 + K_2 e^{-t/\tau}$, $\tau = RC$

For $t = \infty$:

$$V_c(\infty) = K_1 + K_2(0)$$

$$K_1 = V_c(\infty) \text{ — Final voltage at steady state of the capacitor}$$

For $t = 0$:

$$V_c(0) = K_1 + K_2(1)$$

$$K_2 = V_c(0) - V_c(\infty)$$

$$\text{When } t = 5\tau, e^{-t/\tau} \rightarrow 0 \\ \therefore V_c(t) \approx V_c(\infty)$$

At $t = 5\tau$, capacitor have reached steady state.

[Graph of $V(t)$ against t have plotted]

$$\therefore V_c(t) = \underbrace{V_c(\infty)}_{K_1} + \underbrace{[V_c(0) - V_c(\infty)]}_{K_2} e^{-t/\tau}$$

Discharging formulas

no other Voltage source
[$V_c(\infty) = 0$ for simple RC circuit]



$$Q = Q_{\max} (e^{-t/\tau})$$

$$I = I_{\max} (e^{-t/\tau})$$

$$V = V_{\max} (e^{-t/\tau})$$

General formula : $x = x_0 (e^{-t/\tau})$

When $t = \tau$,

$$x = x_0 e^{-1}$$

$$x = 0.37 x_0$$

$\therefore$ After time of τ [$t = \tau = RC$],

Q, V, I will be 37% of its
initial value [Max value]

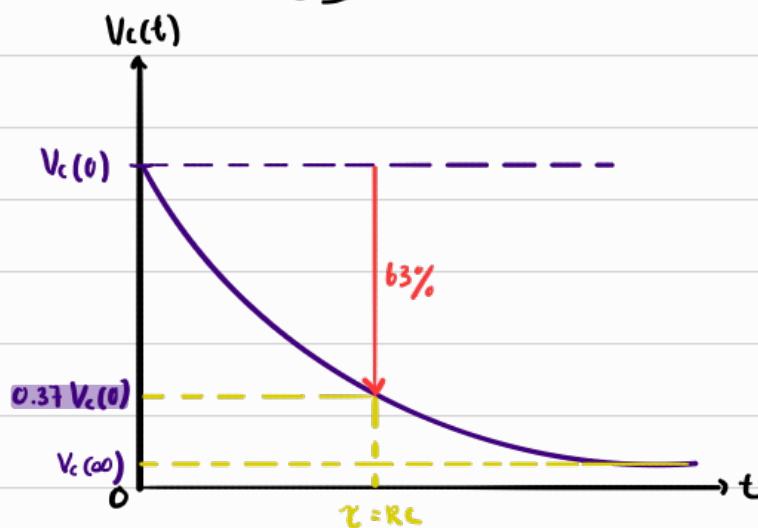
* Note :

At steady state, the capacitor acts as an open circuit, since it's not charging nor discharging, so only a voltage exist across the capacitor, no current

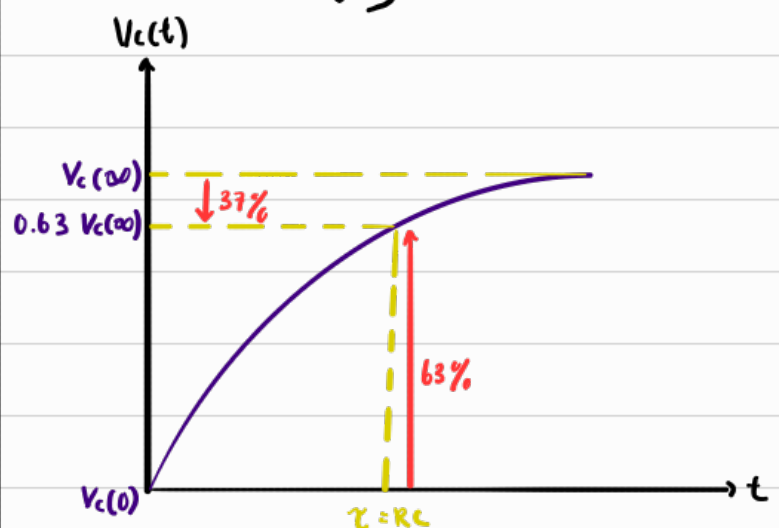
After time of τ [$t = \tau = RC$],

OR Q, V, I have changed by 63% .
[$\Delta V = 0.63$]

Discharging



Charging



Inductor

$$\tau = L/R$$

~ Component that stores energy in a magnetic field

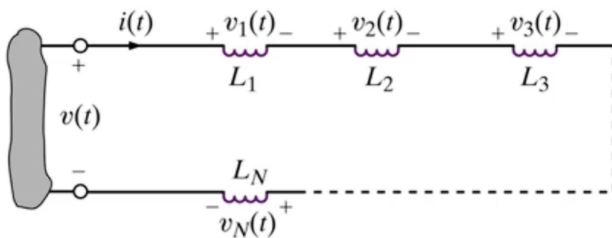
$$V(t) = L \frac{dI(t)}{dt}$$

$$I(t) = I(0) + \frac{1}{L} \int_0^t V(\tau) d\tau$$

Inductance

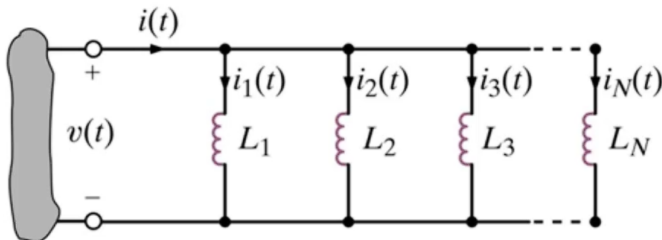
When $\frac{dI}{dt}$ is big, the voltage created by the change in magnetic flux is big [Lenz Law]. This voltage opposes the existing voltage, thus temp. reducing/blocking the incoming current.

Series inductors:



$$L_p = L_1 + L_2 + \dots + L_N$$

Parallel inductors:



$$\frac{1}{L_p} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

Linear Time Invariance (LTI) (Assumption)

Linear ~ ① Output $\propto$ Input

② Superposition is possible

[Feeding 2 inputs at the same time = Sum of each individual fed input]

Time invariant ~ The rules/formula/output is not dependent on when exactly its carried out.

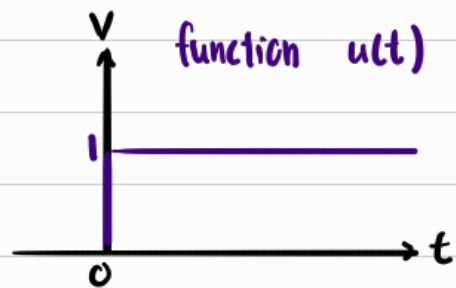
[An input delay of 10s $\rightarrow$ Output delay of 10s]
No change in value.

Step Input

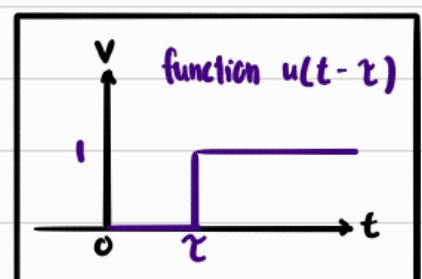
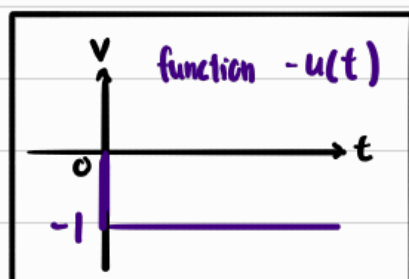
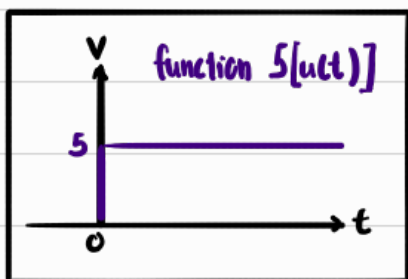
Mathematically defined as

$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

Unit Step Func.
[A 1V]



However we can always manipulate $u(t)$ to get a particular step input function



Example

